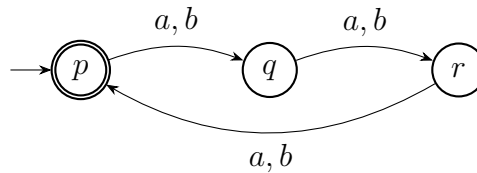


Quiz 1

CPSC 406 — Algorithm Analysis,
Chapman University · Spring 2026
Instructor: Jonathan Weinberger

Name: _____

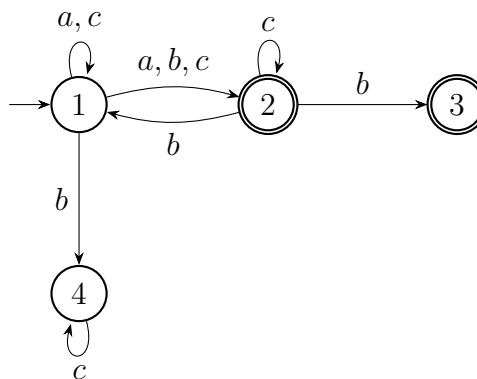
1. (2 points) (a) Consider the following DFA $\mathcal{A}$ over the alphabet $\Sigma = \{a, b\}$:



Which of the following languages is $L(\mathcal{A})$? Tick exactly one.

- ☐ $L_1 = \{ w \in \Sigma^* \mid |w| = 3 \}$
☐ $L_2 = \{ w \in \Sigma^* \mid |w|_a = 3 \}$
☐ $L_3 = \{ w \in \Sigma^* \mid |w| \equiv 0 \pmod{3} \}$
☐ $L_4 = \{ w \in \Sigma^* \mid |w|_a \equiv 0 \pmod{3} \}$
☐ None of these.

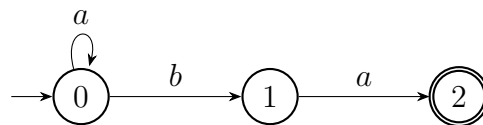
- (b) Consider the following NFA $\mathcal{B}$ over the alphabet $\Sigma = \{a, b, c\}$:



Which of the following words are accepted by $\mathcal{B}$? Tick exactly one.

- ☐ aab
☐ abc
☐ ccc
☐ All of these.
☐ None of these.

2. (2 points) Consider the following NFA $\mathcal{C}$ over the alphabet $\Sigma = \{a, b\}$:



Draw a DFA that recognizes $L(\mathcal{C})$.